Analysis of the Relationship Between World Population Growth and CO2 Emissions

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Abstract

This study investigates the relationship between world population growth and carbon dioxide (CO2) emissions. Using world population and CO2 emissions datasets, the analysis examines population trends, estimates future population, and compares population size with CO2 emissions. Statistical techniques, including correlation analysis and linear regression, are applied to investigate the relationship between these variables. The findings show that the CO2 emission does not depend, mostly on the population. The study highlights the importance of considering economic activity, energy consumption and other factors when interpreting CO2 emission patterns.

1 Introduction

Population growth and CO2 emissions are important global environmental and socioeconomic issues. An increasing population may increase demand for energy, transportation, housing and industrial production. This study investigates whether population growth is associated with increased CO2 emissions.

2 Data and Methodology

The study uses two datasets: a world population dataset and a CO2 emissions dataset. Data cleaning involved checking missing values, duplicates and potential outliers.

Population projections for 2030 were calculated using the exponential growth model:

$$N_t = Pe^{rt}$$

Correlation analysis and linear regression were used to examine the relationship between population and CO2 emissions. Continental emissions were aggregated for 2019.

3 Results and Findings

The ten most populated countries in 2022 were identified and their population trends examined from 1990 to 2022. Figure 1 presents the population ranking, while Figure 2 illustrates changes in population over the study period.

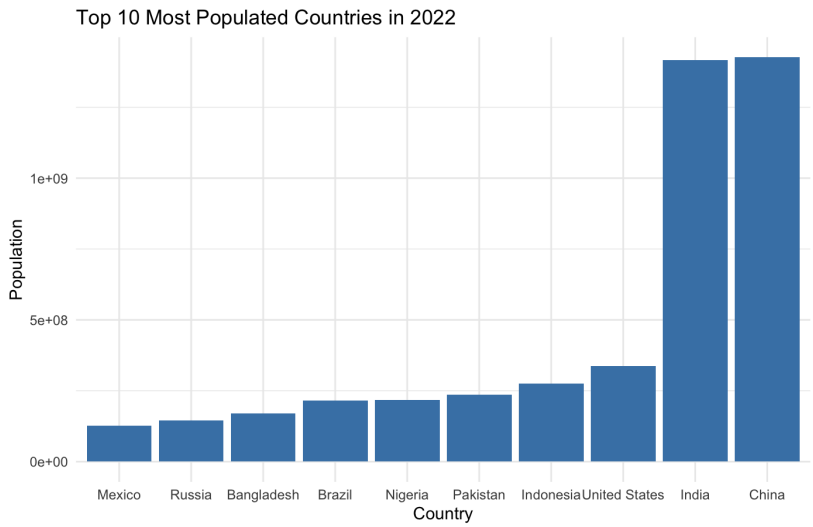


Figure 1: population ranking

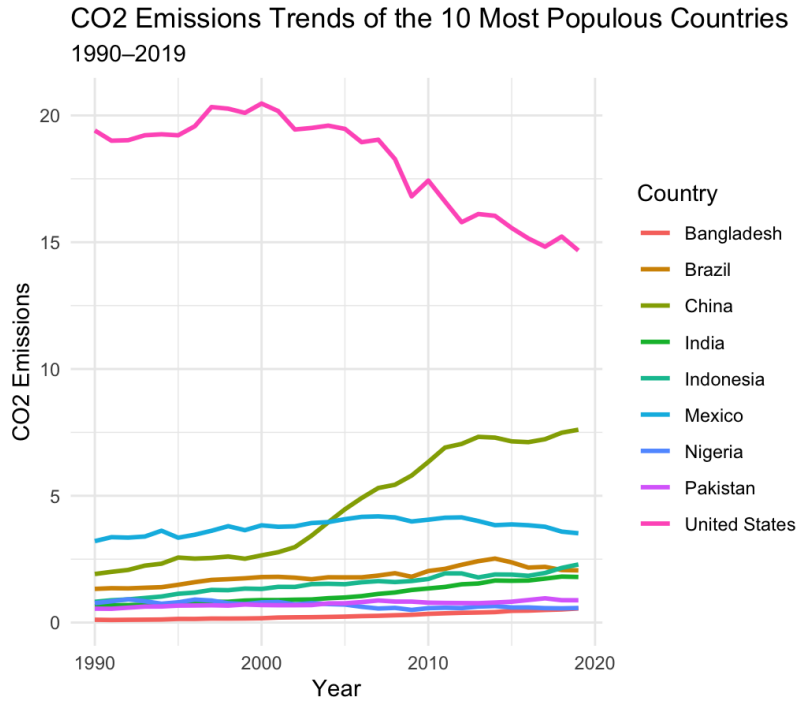


Figure 2: Relationship between 2022 population and 2019 CO_2 emissions.

The relationship between 2022 population and 2019 CO_2 emissions per capita was examined using correlation and linear regression. The analysis revealed a very weak positive correlation ($r = 0.014$) and no statistically significant linear relationship ($p = 0.843$). This suggests that population size alone does not explain differences in CO_2 emissions per person. Finally, average CO_2 emissions per capita were compared across continents in 2019.

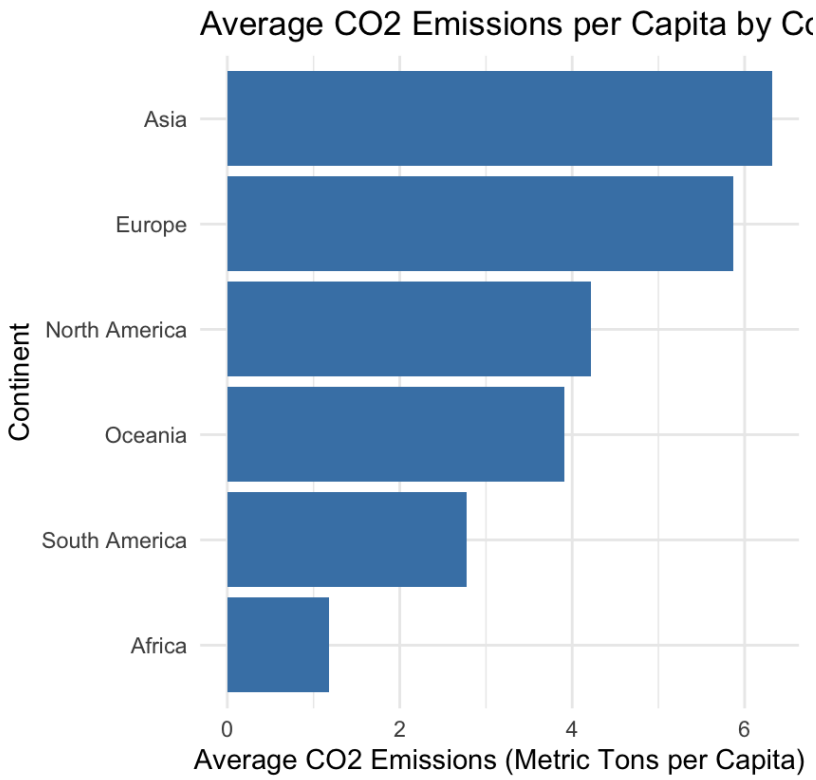


Figure 3: Continents CO2 emission

4 Discussion and Recommendations

The results suggest countries with the most emissions, are not among the most populated ones. Population size alone may not explain CO2 emissions because industrialization, energy sources, economic activity and consumption patterns also influence emissions.

Countries should promote energy efficiency, renewable energy and sustainable development.

5 Conclusion

This study examined population growth and CO2 emissions using descriptive and statistical methods. The findings indicate that population size alone is insufficient to explain differences in CO_2 emissions per capita. Other factors, including energy consumption, industrialization, economic development and energy sources, should also be considered. Future research could include GDP, energy consumption and CO2 emissions per capita to provide a more comprehensive analysis.